Analog to Digital Conversion

以聲音(語音)為例 (粗淺內容)

江振宇 副教授
國立臺北大學
通訊工程學系
20200913

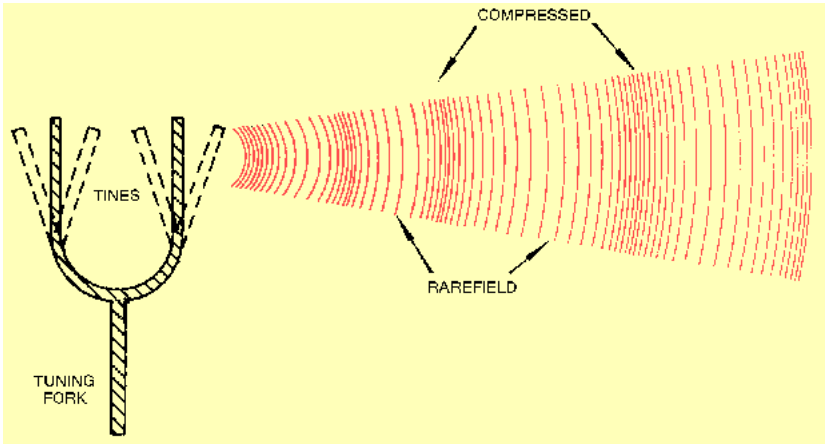
How computer/machine listen?

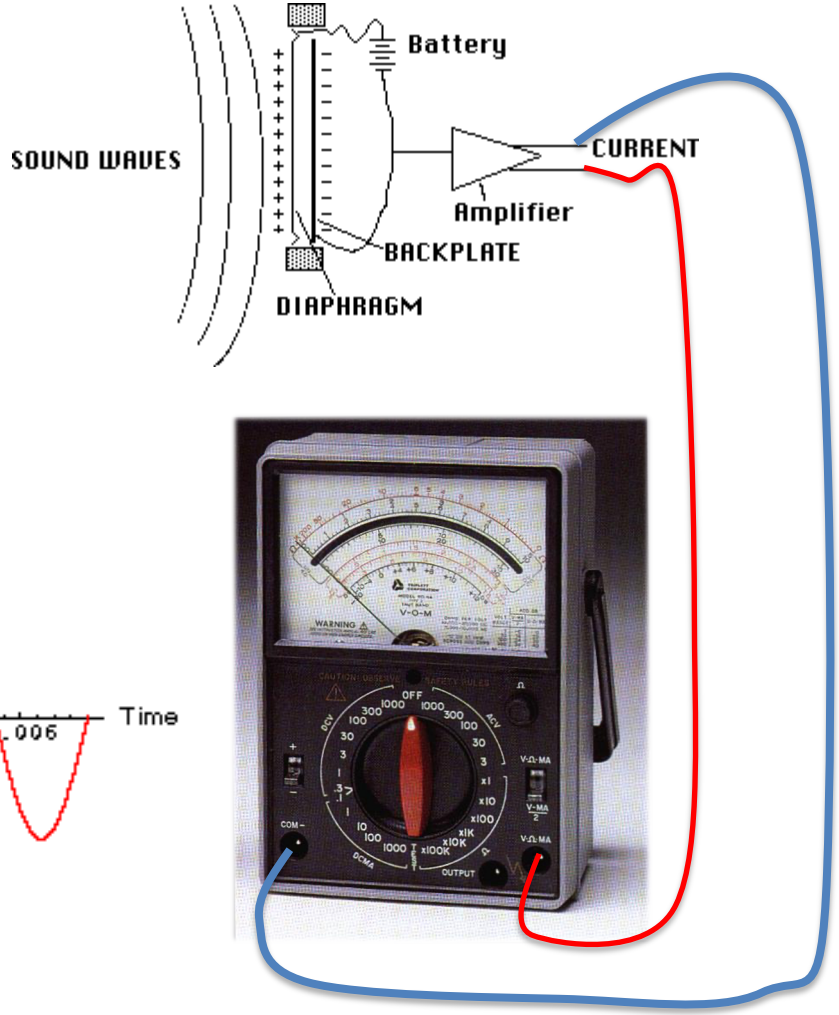
- Microphone as an ear!

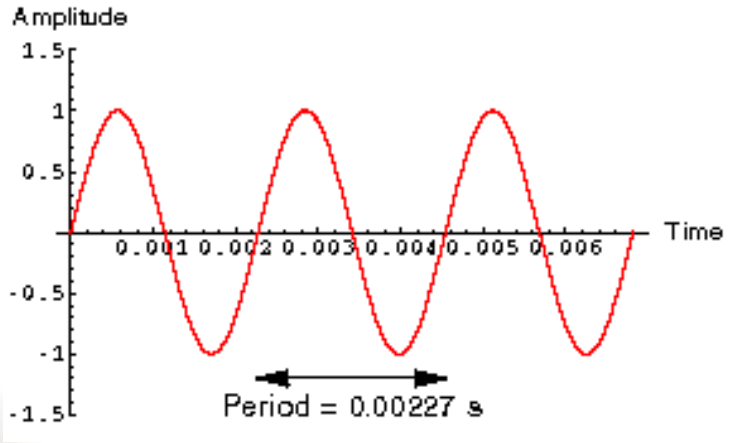




How microphone works?







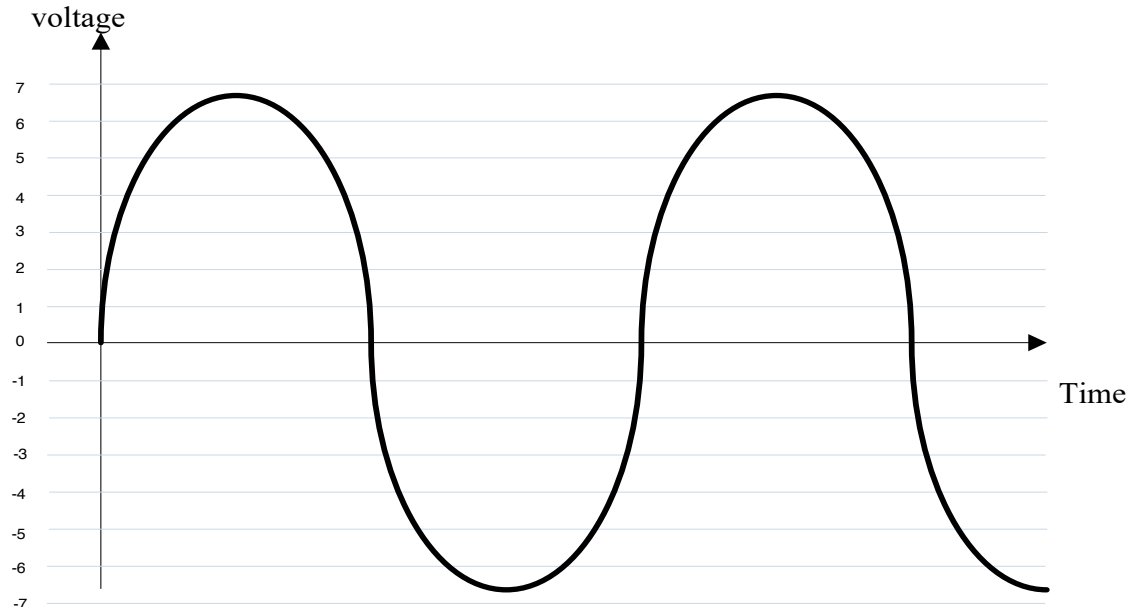
waveform



http://artsites.ucsc.edu/EMS/music/tech_background/te-20/teces_20.html

How computer/machine stores sounds?

Continuous signal
- **infinite** possible point
in time and amplitude

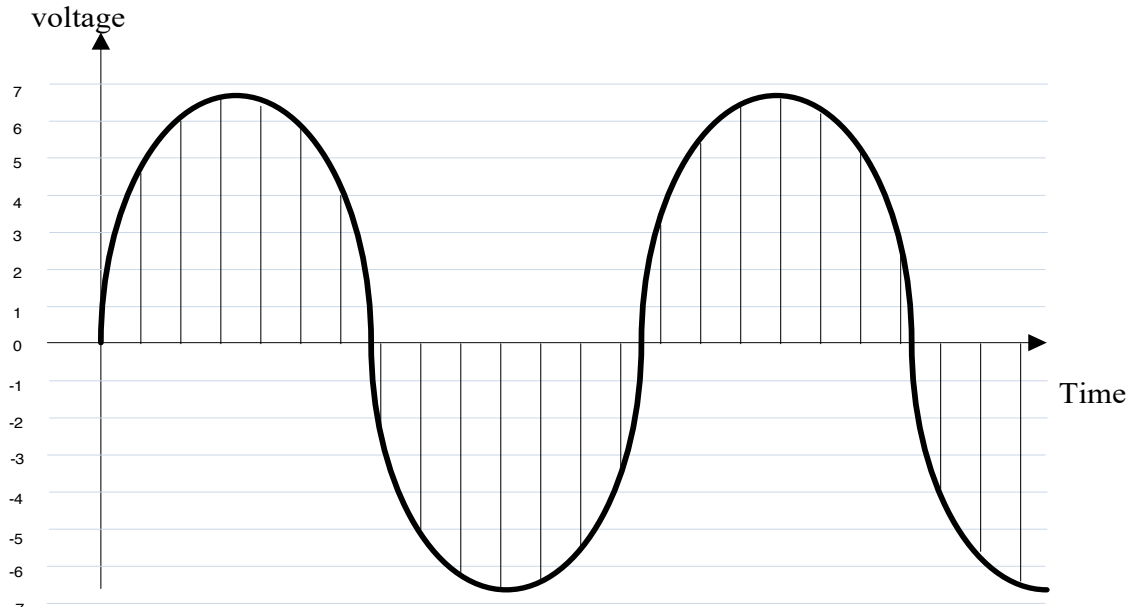


How computer/machine stores sounds?

Continuous signal
- **infinite** possible point in time and amplitude

→ Sampling every *T* seconds →

Discrete signal - **infinite** possible point in amplitude, but **finite** point in time

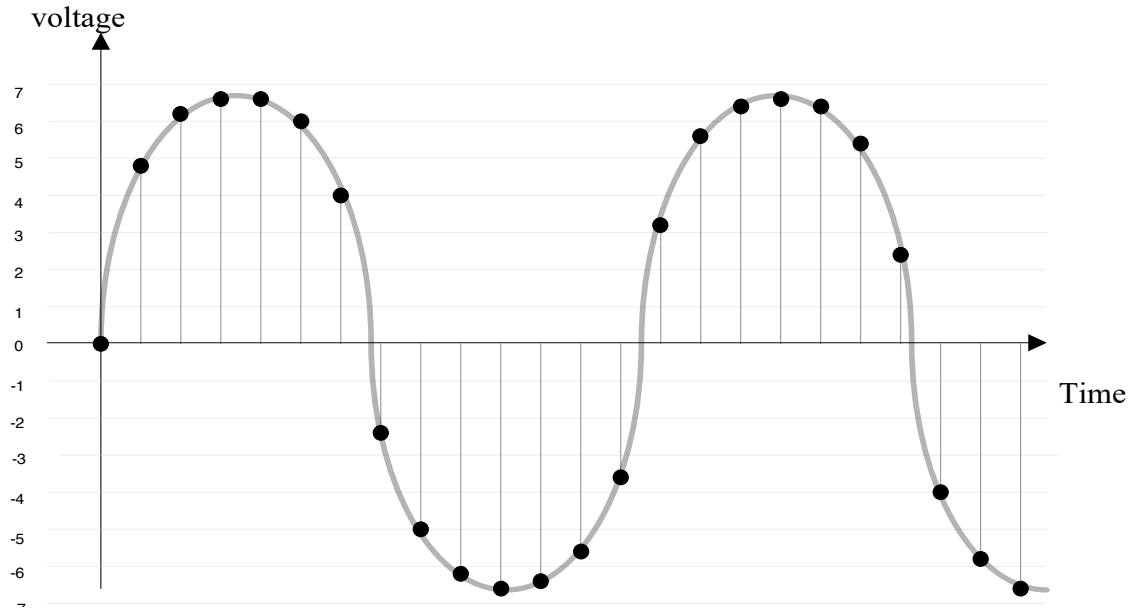


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How computer/machine stores sounds?

Continuous signal - **infinite** possible point in time and amplitude

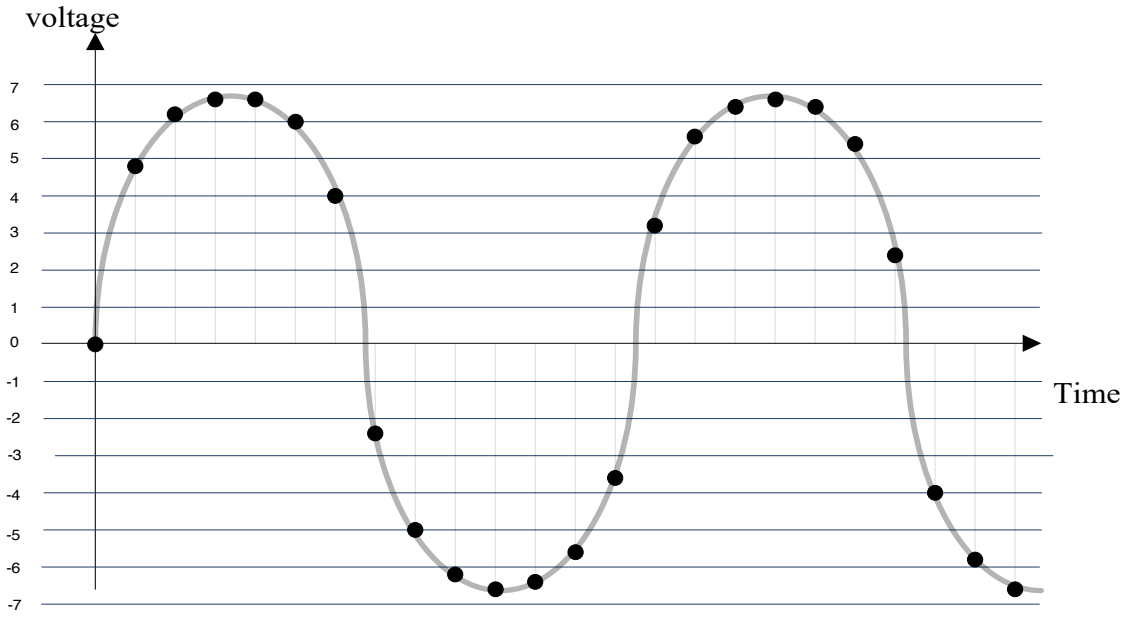
→ Sampling every *T* seconds

→ Discrete signal - **infinite** possible point in amplitude, but **finite** point in time

→ Quantizing in *M* levels

↓

Discrete signal - **finite** possible point in time and amplitude



How computer/machine stores sounds?

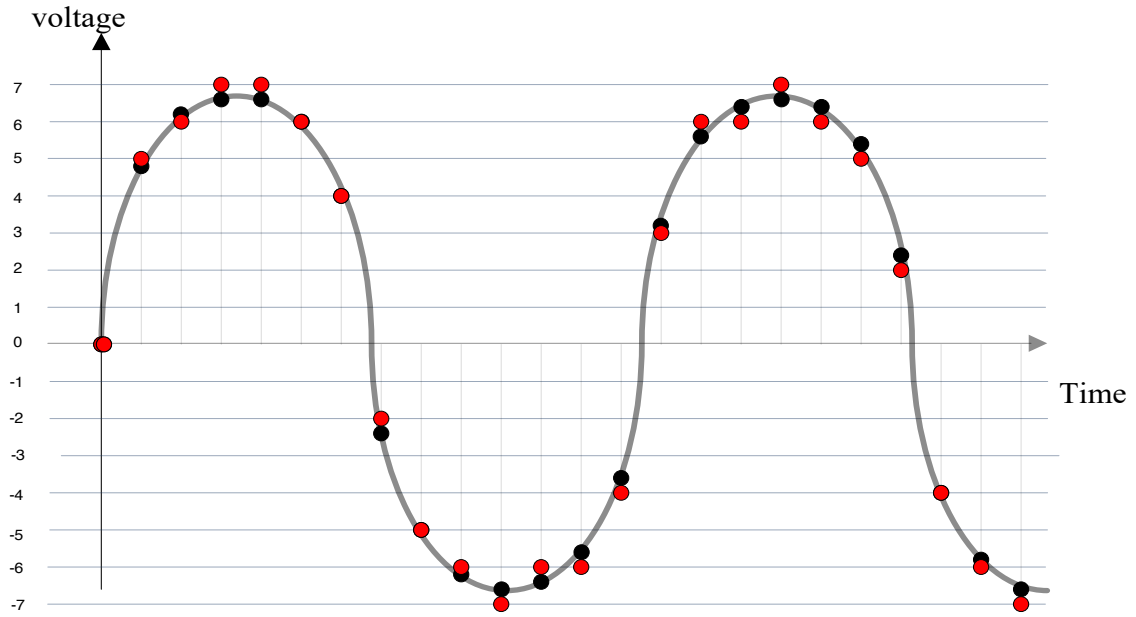
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Discrete signal - **finite** possible point in time and amplitude



How computer/machine stores sounds?

Continuous signal - **infinite** possible point in time and amplitude

→ Sampling every *T* seconds

→ Discrete signal - **infinite** possible point in amplitude, but **finite** point in time

→ Quantizing in *M* levels

→ Discrete signal - **finite** possible point in time and amplitude

→ Encoding in binary (1/0)

→ Digital signal - binary representation in a media

$[0111]_2 \rightarrow 7 = (-1)^0 \times (1\times 2^2 + 1\times 2^1 + 1\times 2^0)$

$[0110]_2 \rightarrow 6 = (-1)^0 \times (1\times 2^2 + 1\times 2^1 + 0\times 2^0)$

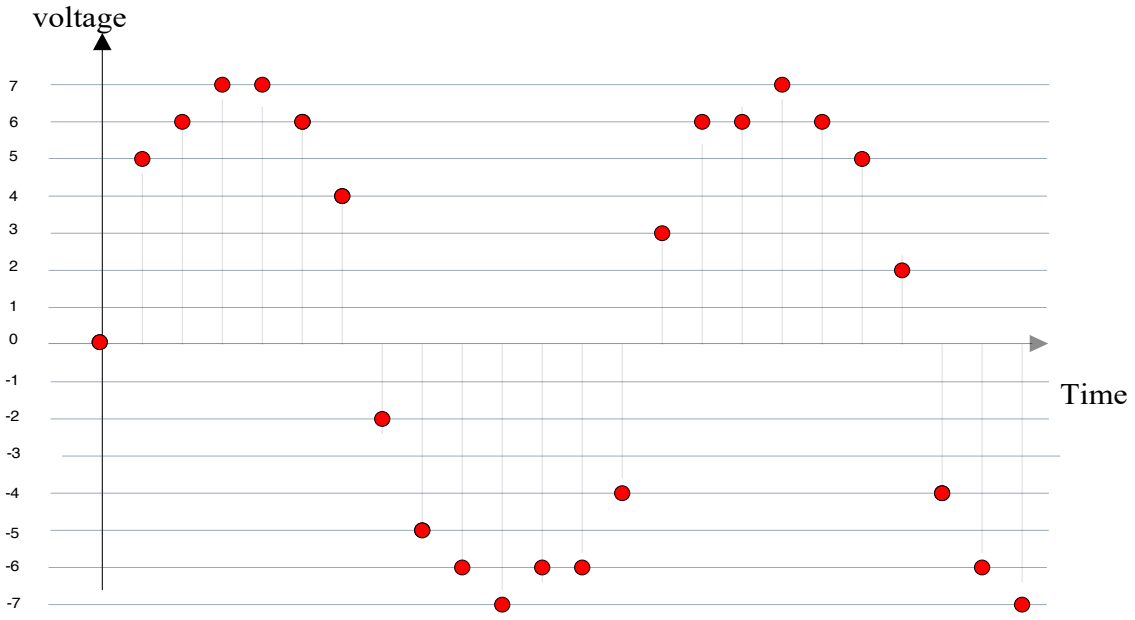
$[0010]_2 \rightarrow 2 = (-1)^0 \times (0\times 2^2 + 1\times 2^1 + 0\times 2^0)$

$[0000]_2 \rightarrow 0 = (-1)^0 \times (0\times 2^2 + 0\times 2^1 + 0\times 2^0)$

$[1010]_2 \rightarrow -2 = (-1)^1 \times (0\times 2^2 + 1\times 2^1 + 0\times 2^0)$

$[1110]_2 \rightarrow -6 = (-1)^1 \times (1\times 2^2 + 1\times 2^1 + 0\times 2^0)$

$[1111]_2 \rightarrow -7 = (-1)^1 \times (1\times 2^2 + 1\times 2^1 + 1\times 2^0)$



How computer/machine stores sounds?

Continuous signal - **infinite** possible point in time and amplitude

→ Sampling every *T* seconds

→ Discrete signal - **infinite** possible point in amplitude, but **finite** point in time

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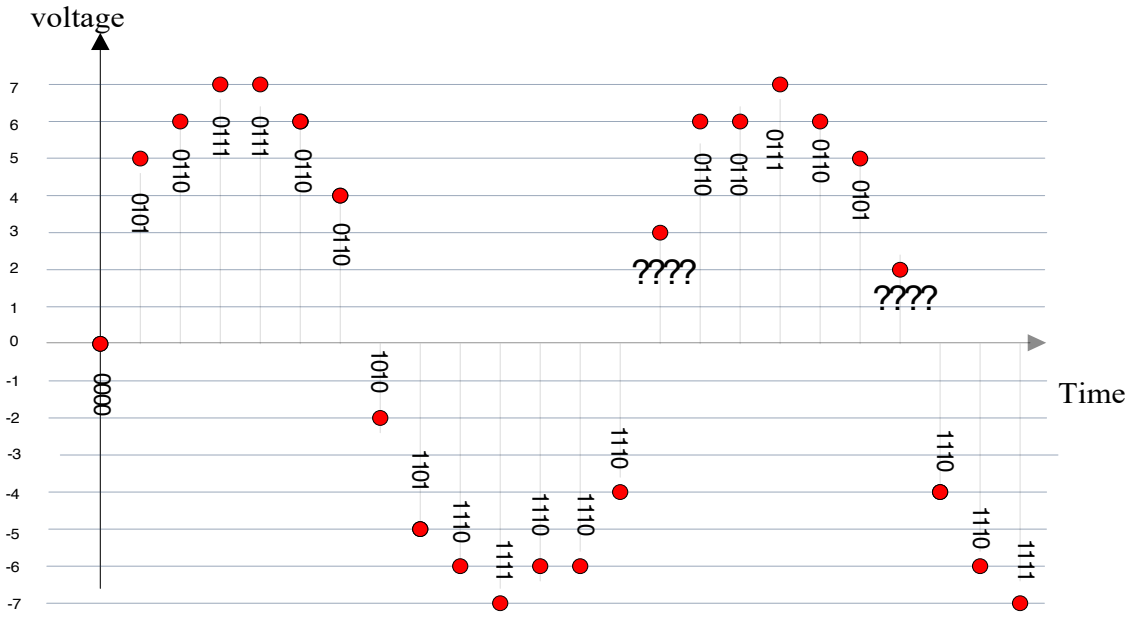
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How computer/machine stores sounds?

Continuous signal - **infinite** possible point in time and amplitude

→ Sampling every *T* seconds

→ Discrete signal - **infinite** possible point in amplitude, but **finite** point in time

→ Quantizing in *M* levels

↓

Discrete signal - **finite** possible point in time and amplitude

← Encoding in binary (1/0)

← Digital signal - binary representation in a media



0000 0101 0110 0111 0111 0110 0100 1010 1101 … ???? ……..????……..

0 4 6 7 7 6 4 -1 -5 3 2

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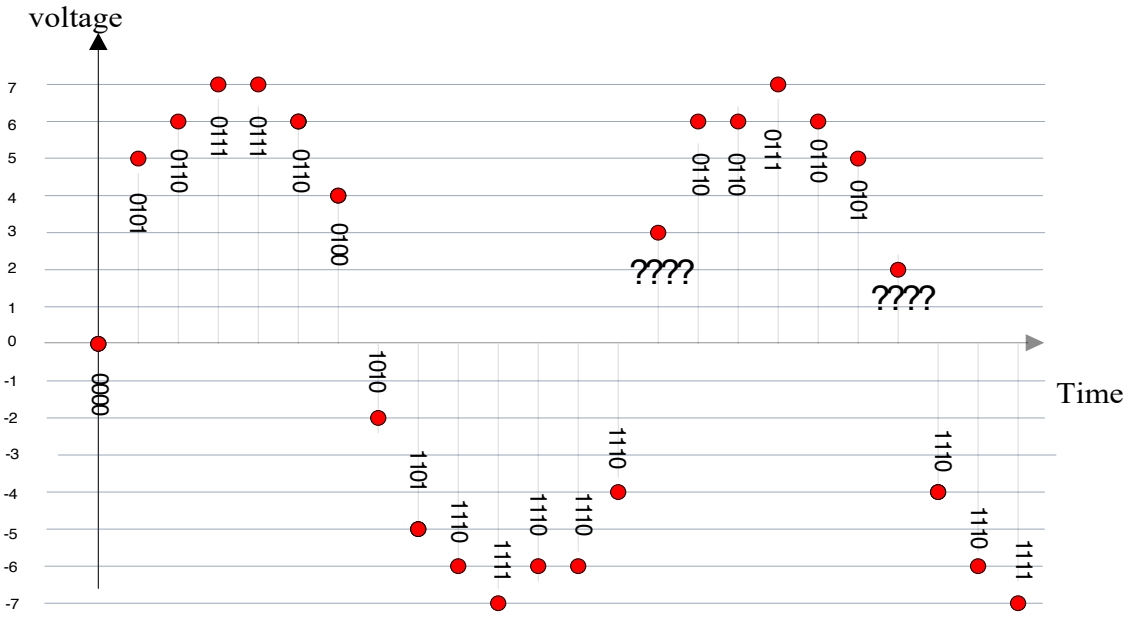
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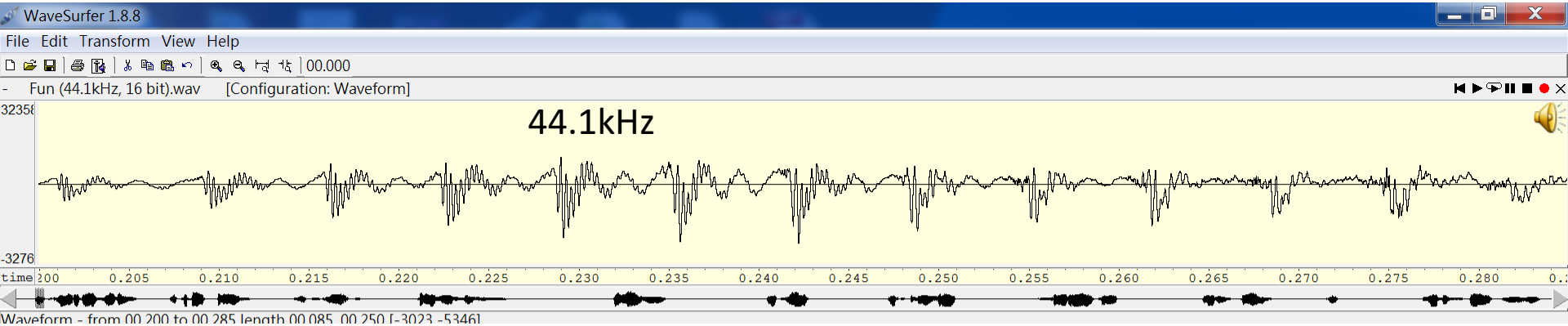
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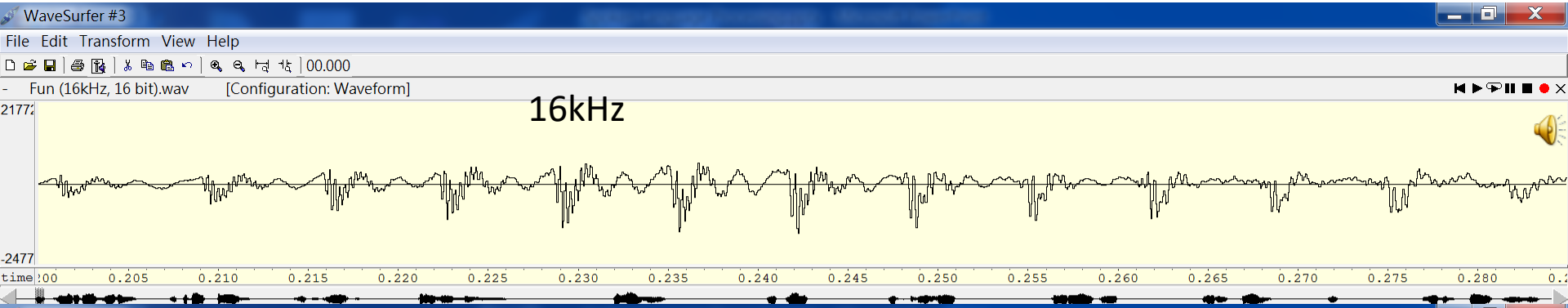


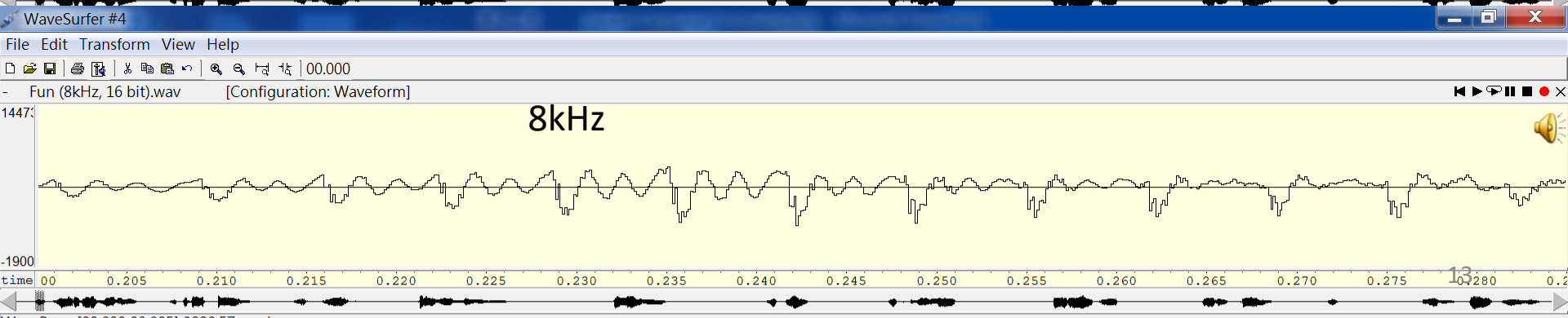
Frequent-used Waveform Format

- 44.1KHz, 16 bits
  - CD
- 16kHz, 16 bits
  - speech recognition
- 8kHz (sampling rate), 16 bits resolution
  - telephone

‘it’ spoken by SpongeBob

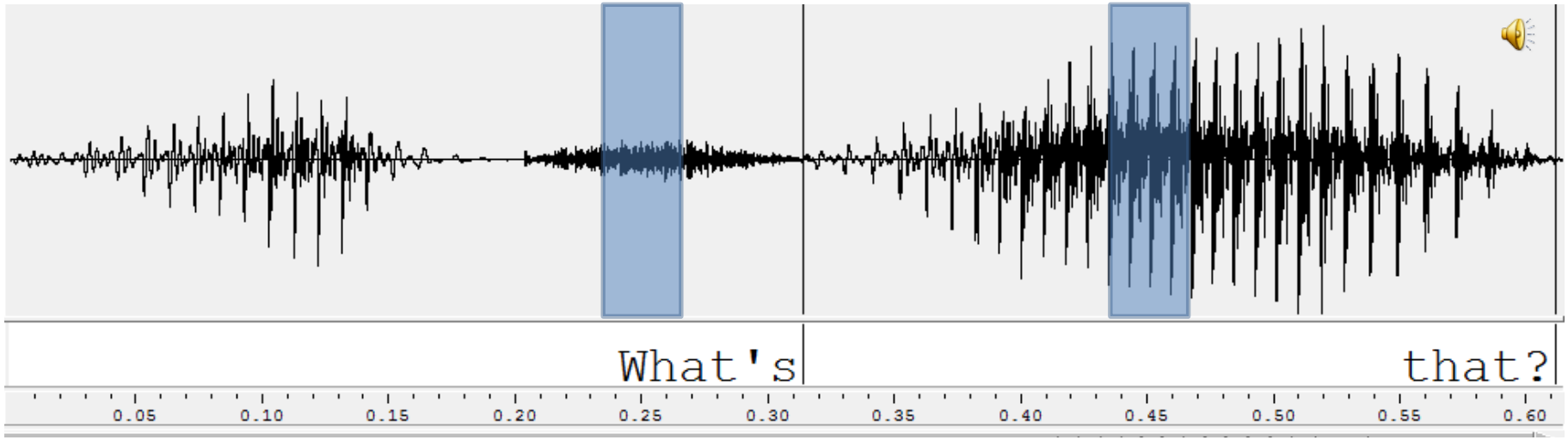


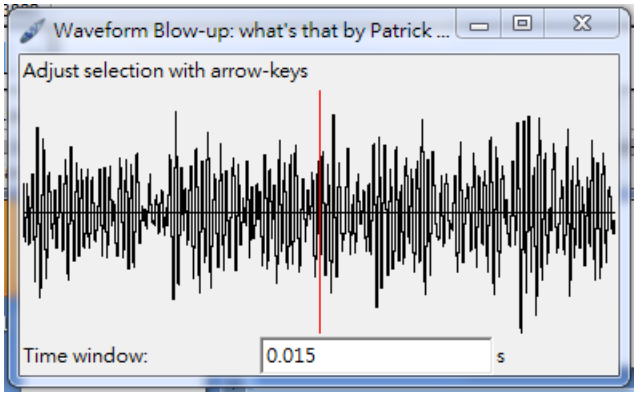


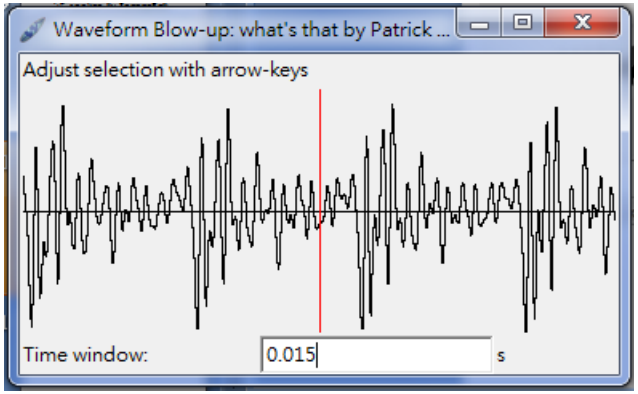


Analysis on Speech Signal

- Speech varies over time, but may be stationary in a short time window.





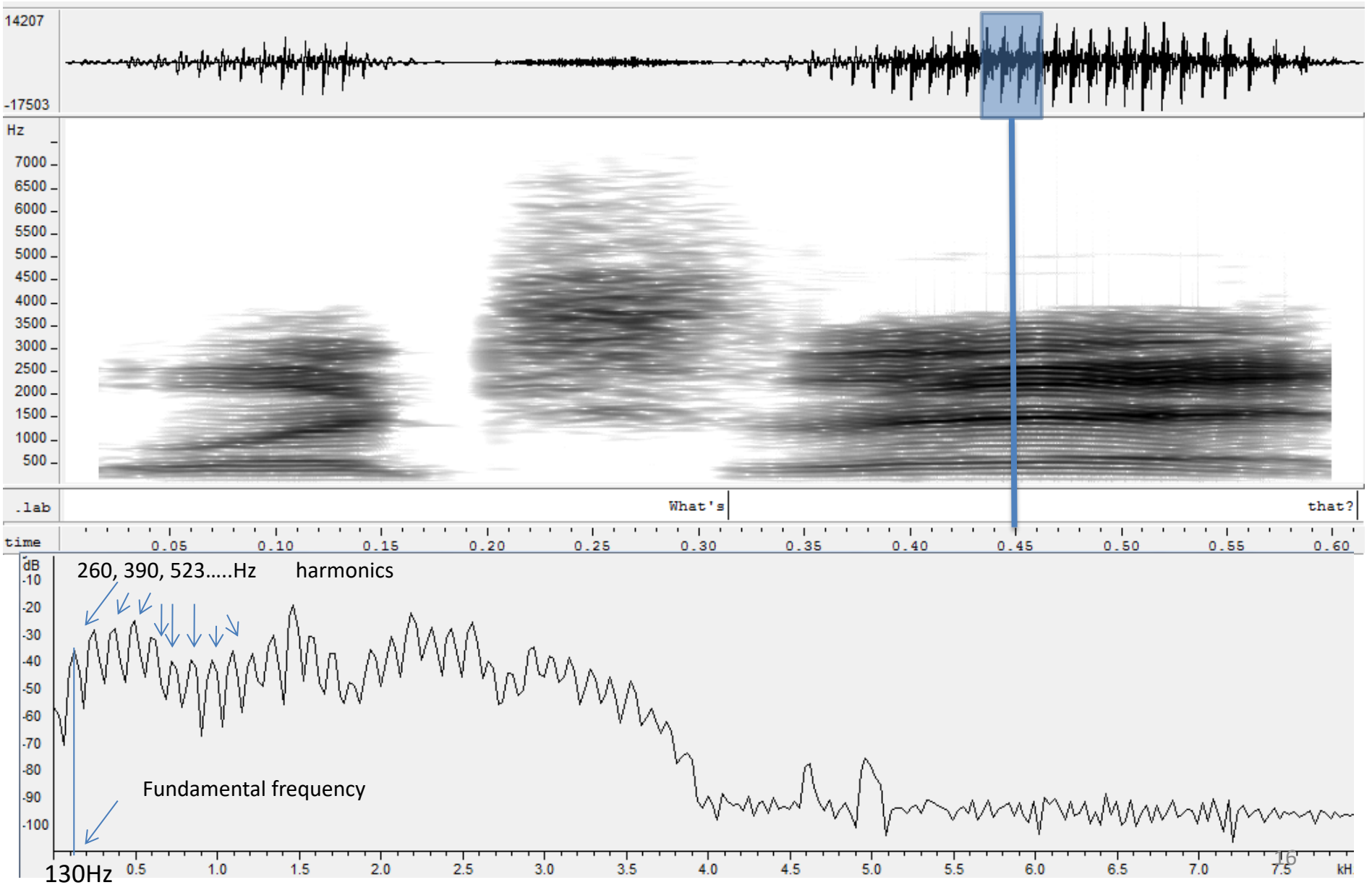


Decomposition of a speech signal

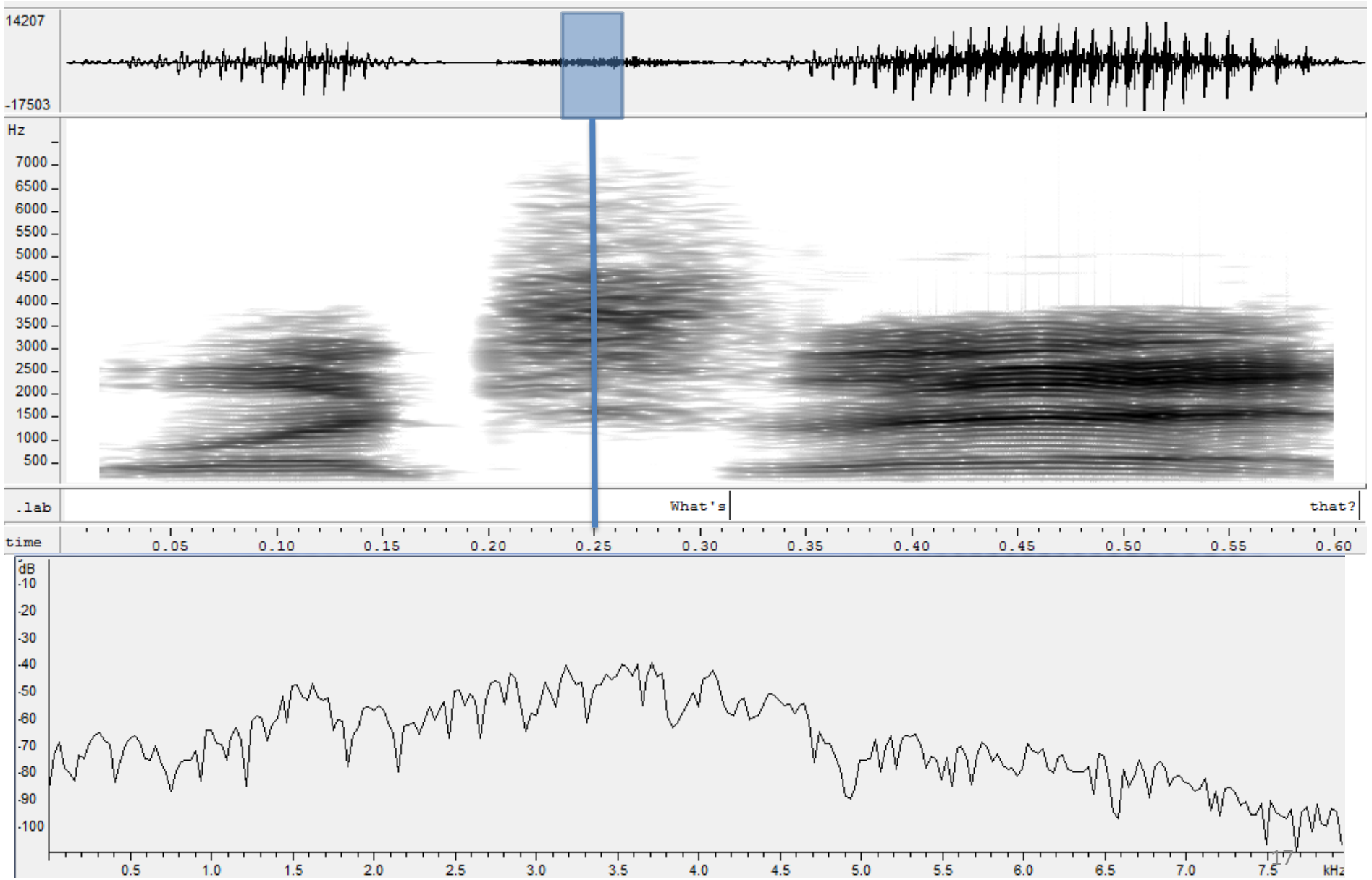
- A signal can be decomposed into several components, including
  - Fundamental frequency f → pitch
  - Harmonics (2*f, 3*f,….N*f)
- Examples of ‘ts’ and ‘a’.

Idea of Fourier Transform!

Spectrogram

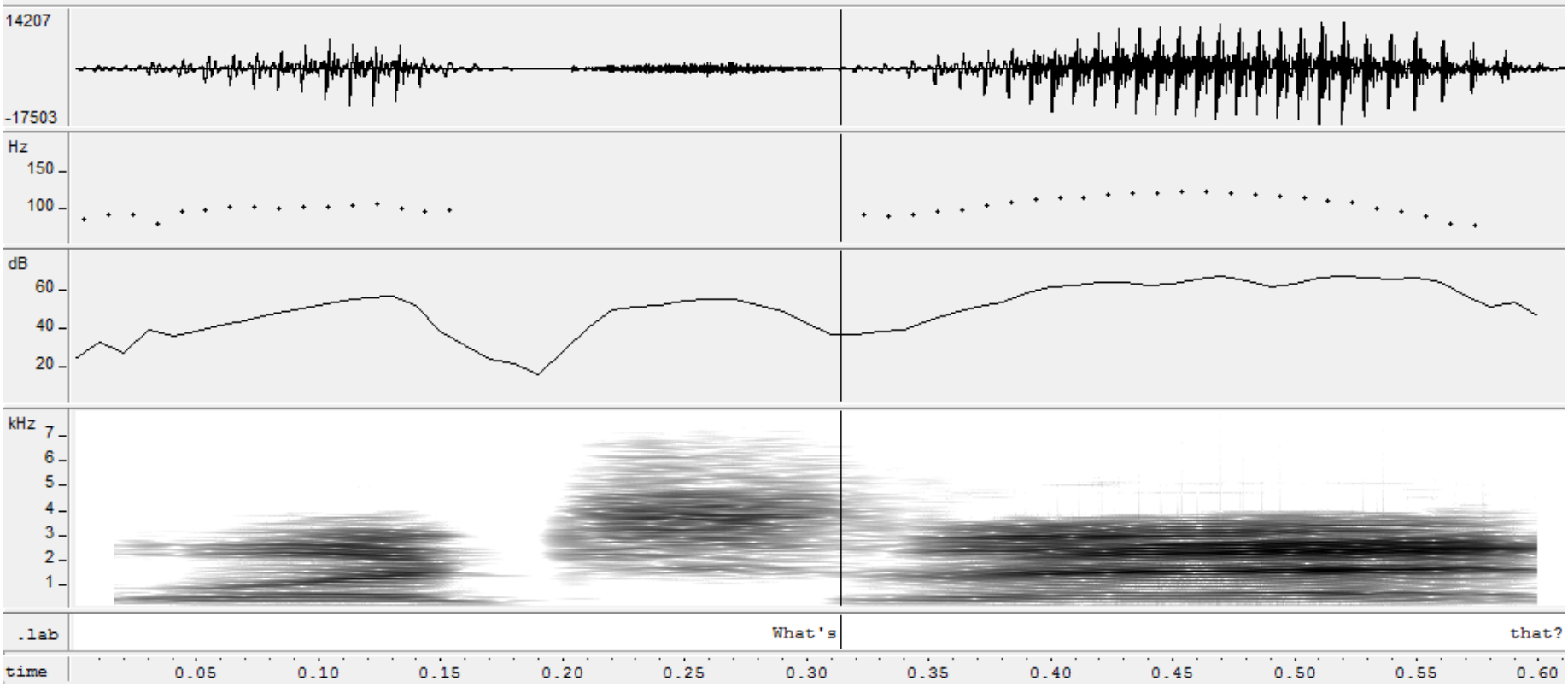


Spectrogram



Recall – elements of speech

- Pitch, Duration, Intensity, Timbre



What distinguishes types of sound?

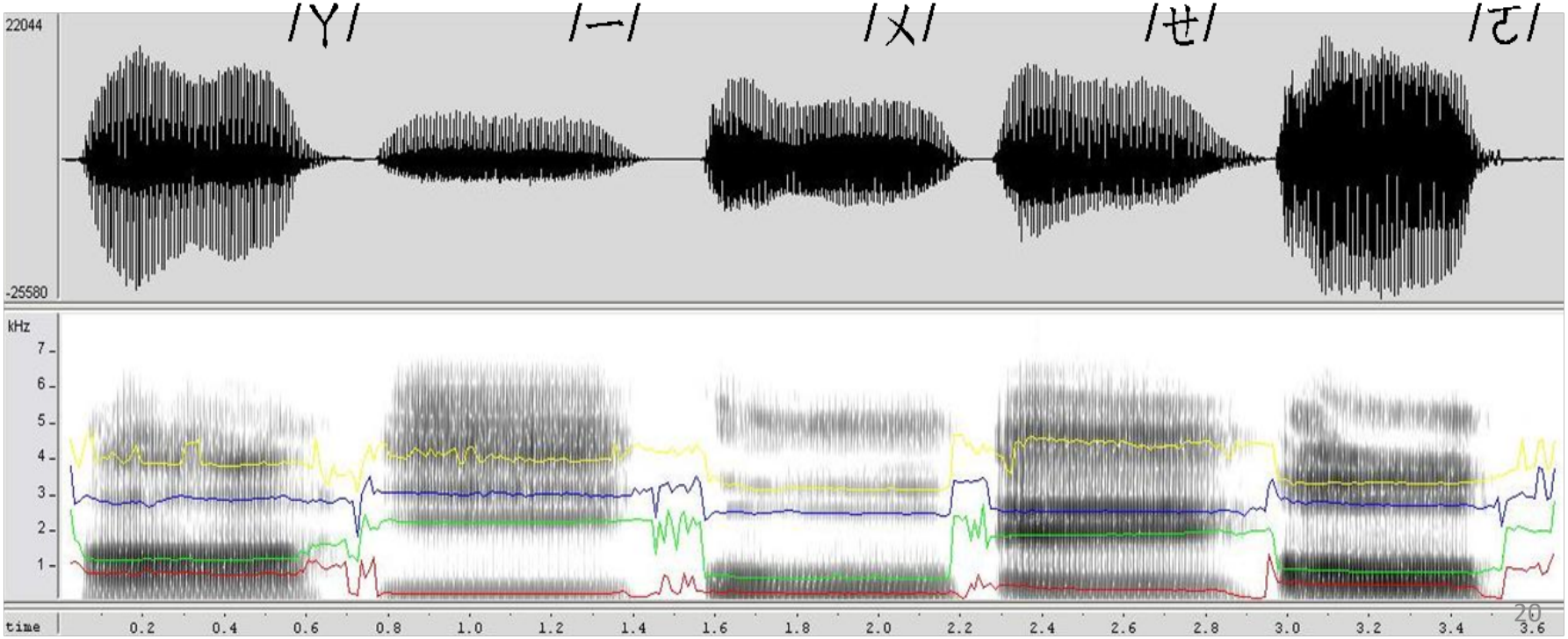
- In music → timbre

Subjective	*Objective*
Tonal character, usually pitched	Periodic sound
Noisy, with or without some tonal character, including rustle noise	Noise, including random pulses characterized by the rustle time (the mean interval between pulses)
Coloration	Spectral envelope
Beginning/ending	Physical rise and decay time
Coloration glide or formant glide	Change of spectral envelope
Microintonation	Small change (one up and down) in frequency
Vibrato	Frequency modulation
Tremolo	Amplitude modulation
Attack	Prefix
Final sound	Suffix

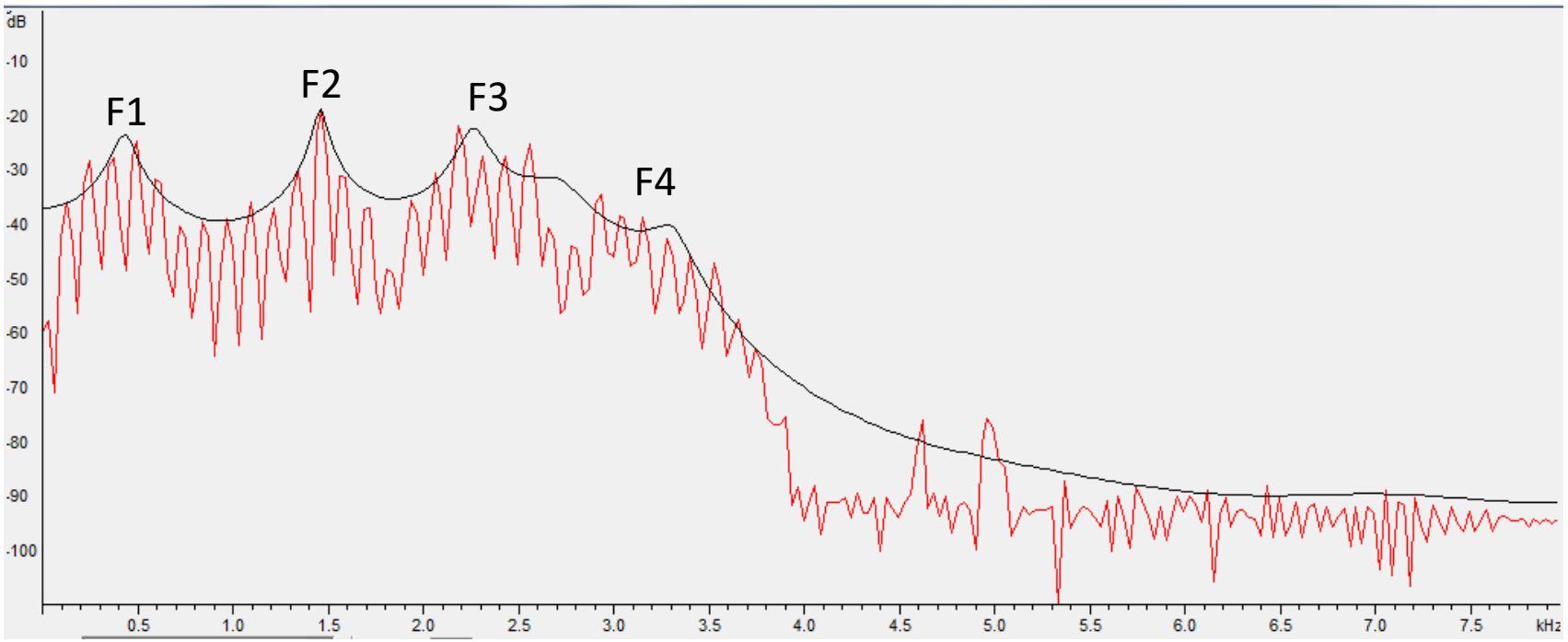
Reference: Xuedong Huang, Alex Acero, Hsiao-Wuen Hon, and Raj Reddy. 2001. Spoken Language Processing: A Guide to Theory, Algorithm, and System Development (1st. ed.). Prentice Hall PTR, USA.

What distinguishes types of sound?

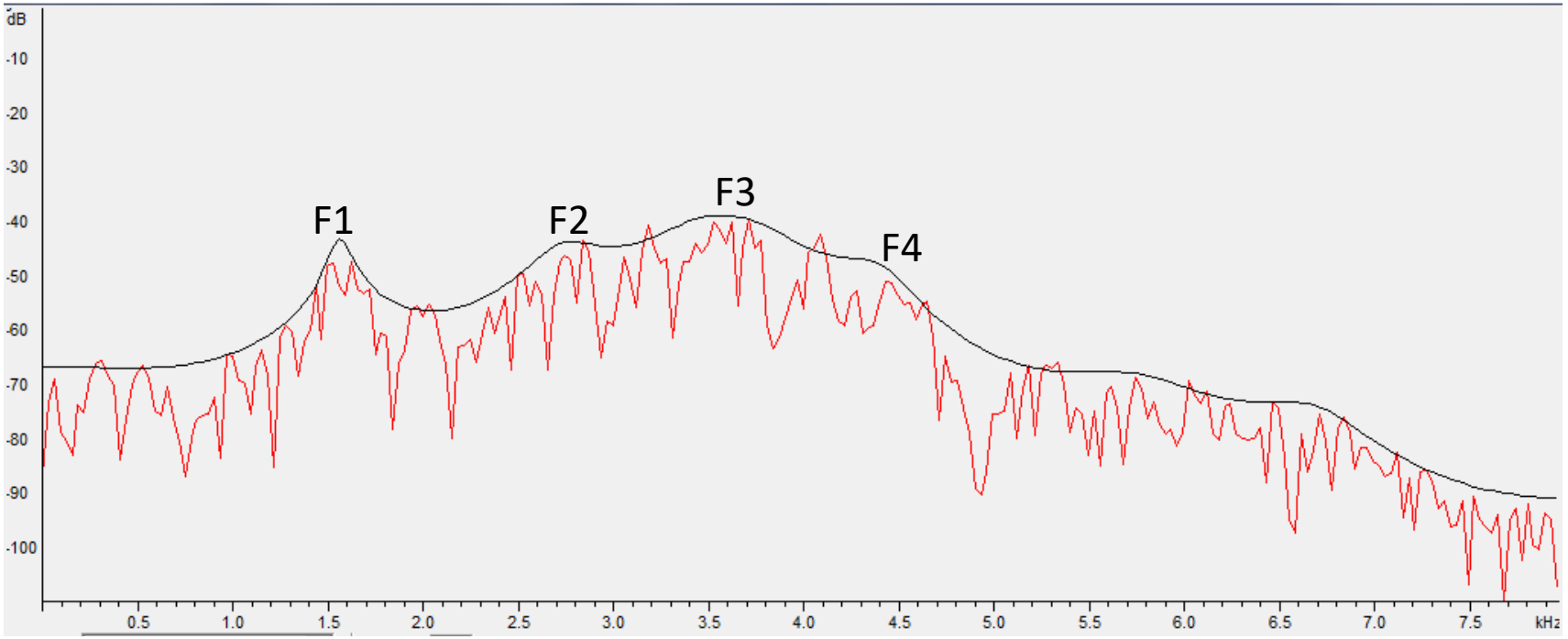
- In speech→ formant
  - Peaks (F1,F2,…) in the spectrum of a sound caused by acoustic resonance.
  - In phonetics, the formant refers to sounds produced by the vocal tract.



‘a’


dB
-10
-20
-30
-40
-50
-60
-70
-80
-90
-100
F1
F2
F3
F4
0.5
1.0
1.5
2.0
2.5
3.0
3.5
4.0
4.5
5.0
5.5
6.0
6.5
7.0
7.5
kHz

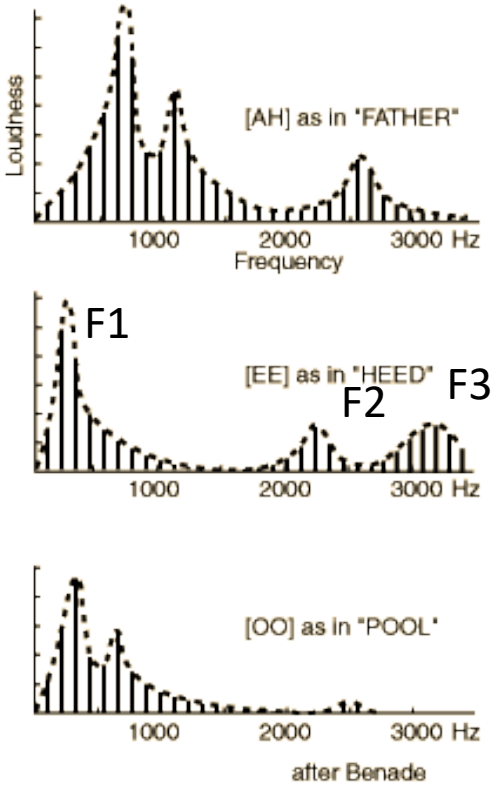
‘ts’

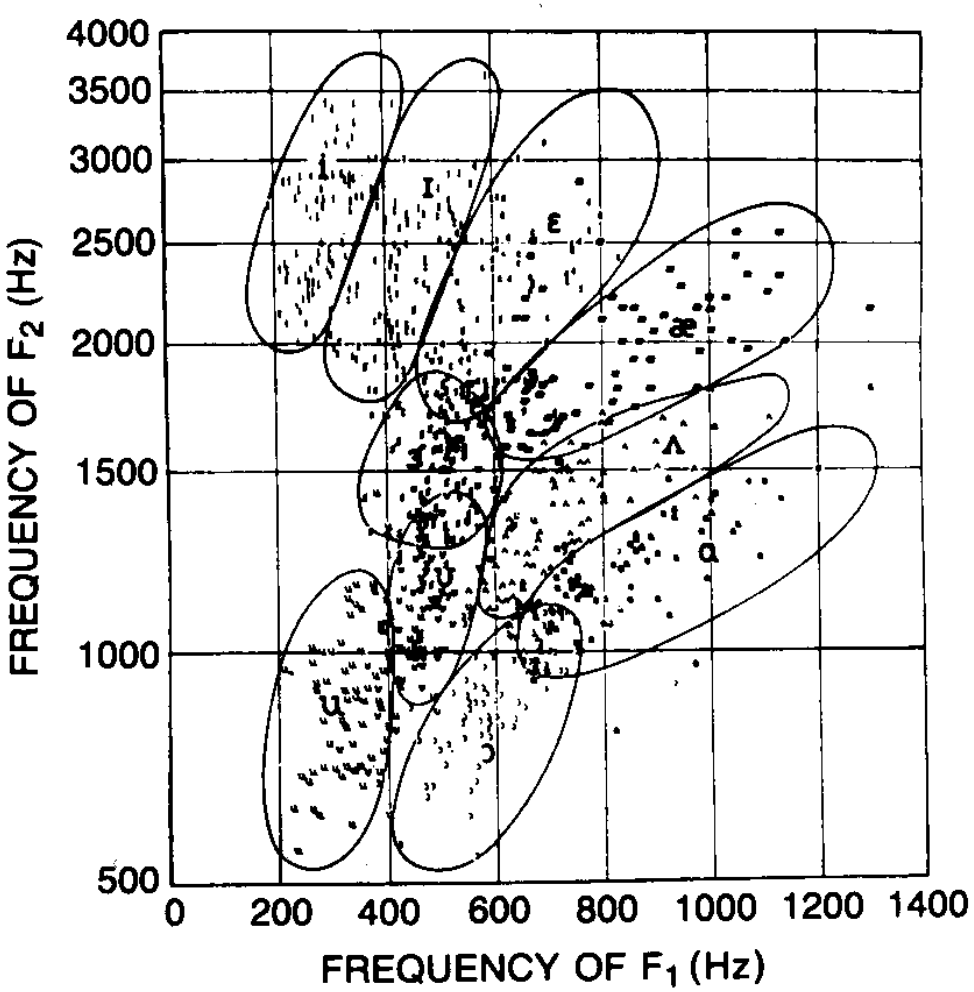

dB
-10
-20
-30
-40
-50
-60
-70
-80
-90
-100
F1
F2
F3
F4
0.5
1.0
1.5
2.0
2.5
3.0
3.5
4.0
4.5
5.0
5.5
6.0
6.5
7.0
7.5
kHz

Formant

- Distinctive feature for automatic speech recognition by machine

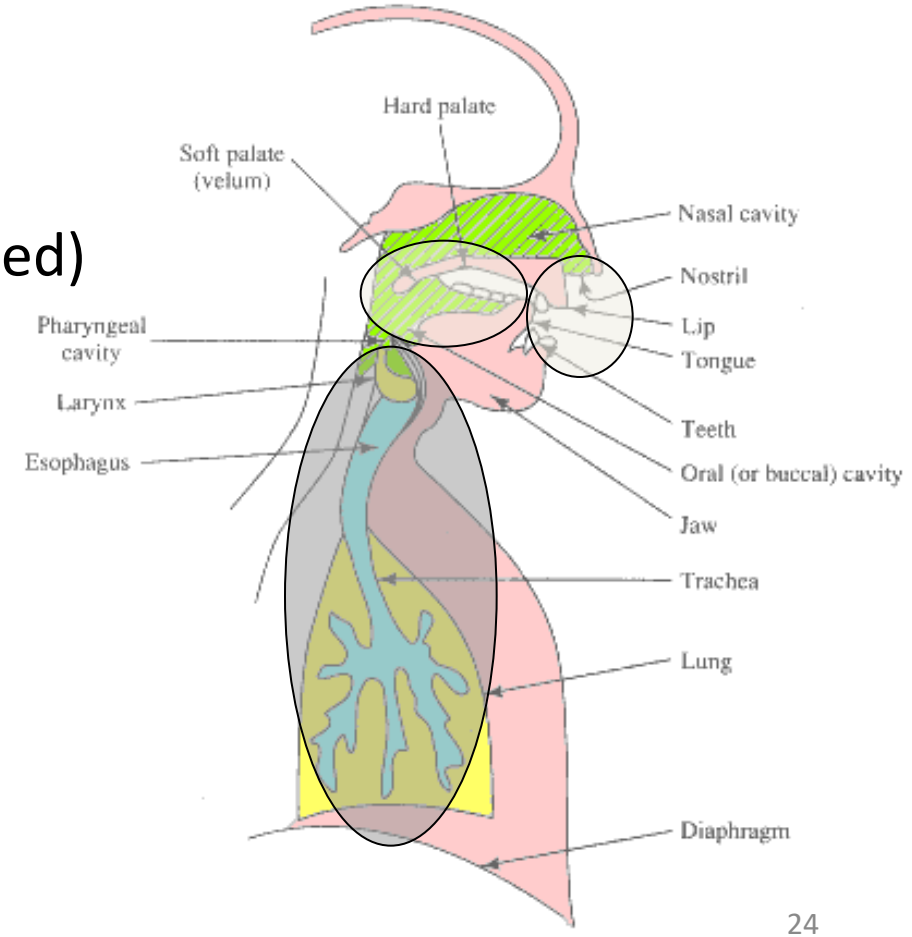






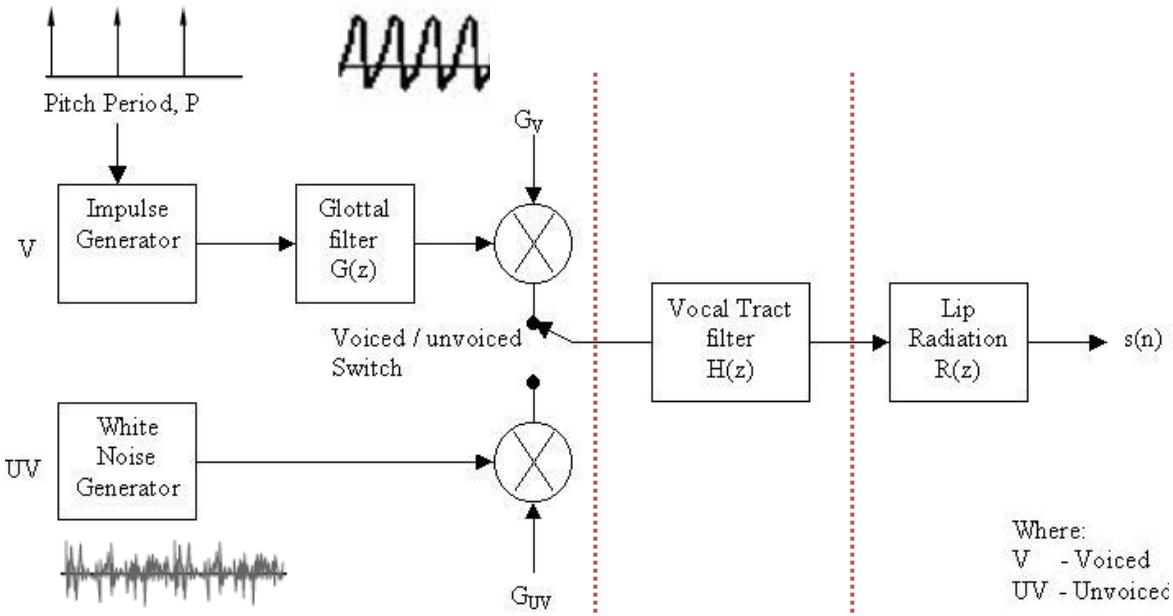
Speech Production

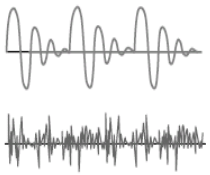
- How speech generated?
  - Excitation Source
    - Lung
    - Larynx (voiced or unvoiced)
  - Propagation
    - Vocal tract (timbre)
    - Nasal
  - Radiation
    - Lips to free space

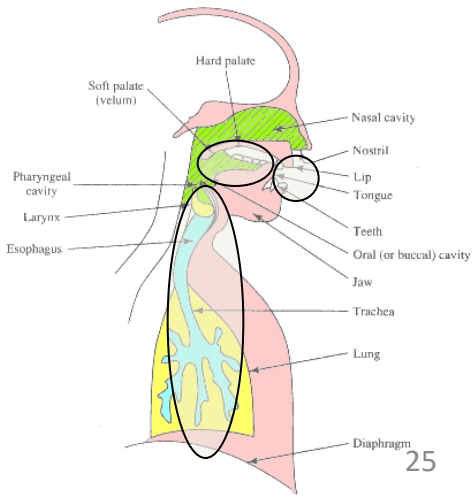


Speech Production

- Digital model for speech production







Excitation Source

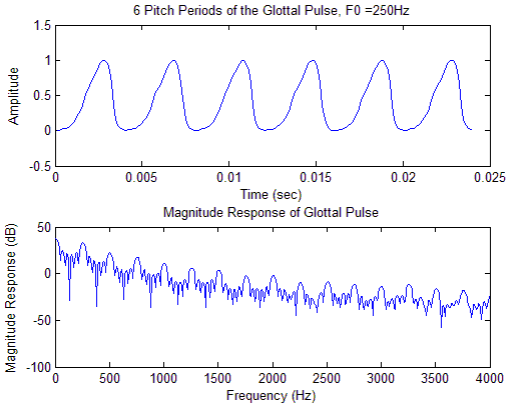
Propagation

Radiation

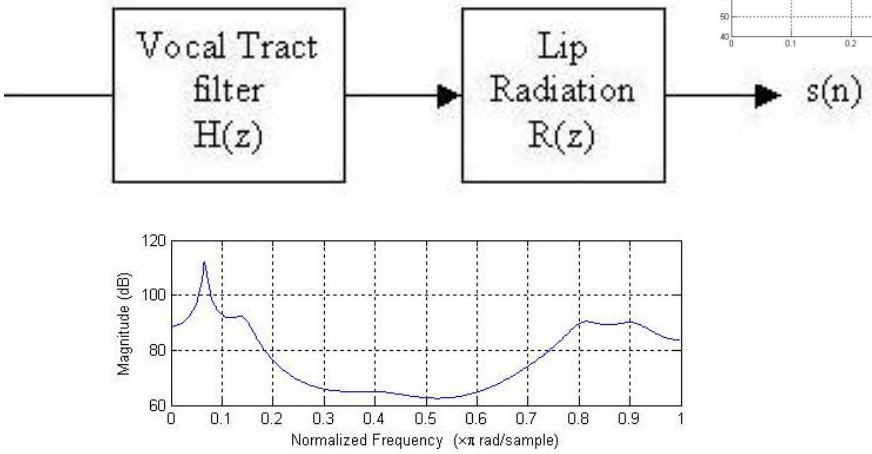
Speech Production

- Source-Filter Model and the corresponding spectra

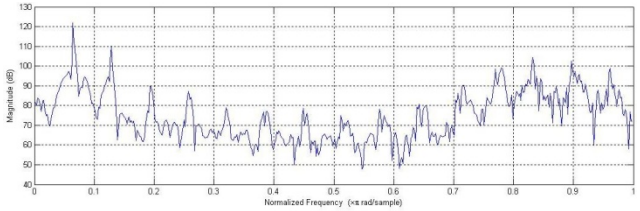
Glottal Pulse



Vocal Tract



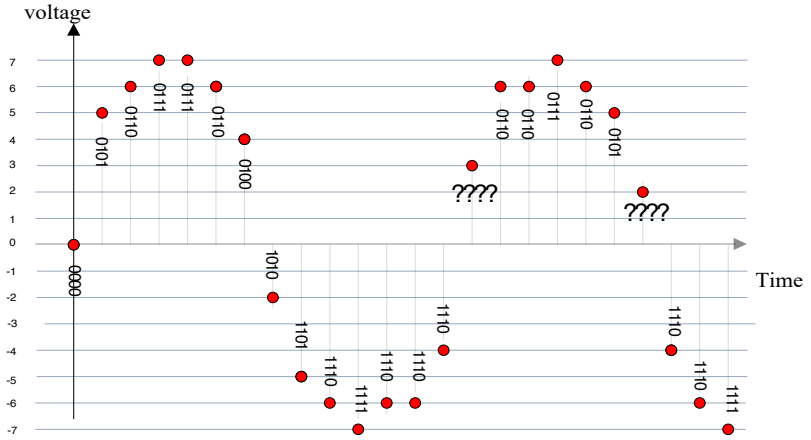
Speech Signal



Speech Coding

- PCM in CD, telephone network



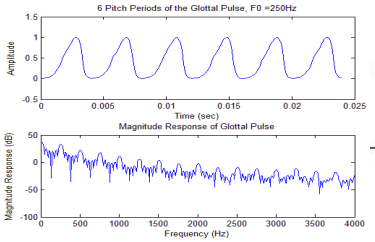


- CELP in GSM, Skype

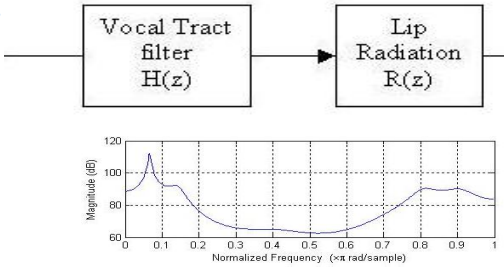




Glottal Pulse



Vocal Tract



Speech Signal

